

Goals

Core Questions that need to be answered:

- **Inputs:** What sort of data goes in?
- **Output Schema:** What potential structured format comes out (eg: JSON)?
- **Failure Modes:** Where does the LLM hallucinate or break down?
- **Validation Plan:** How shall we guardrail and test it?

Profile Agent Design Focus:

- **Objective:** Build a structured intake system
- **Core Logic:** Needs to capture total wealth, combining financial assets with human capital (the present value of future earnings).
- **Key Variables:** Career type, income stability, industry-specific risk exposure, current holdings, investment horizon, and risk tolerance.
- **Outputs:** Comprehensive total wealth assessment to pass down the pipeline.

Research Agent Design Focus:

- **Objective:** Scan macroeconomic data
- **Core Logic:** It needs to work with FRED data and run a regime-matching model.
- **Key Variables:** Yield curves, credit spreads, unemployment rates, and CPI
- **Outputs:** Definitive macro regime assessment that compares current data against historical analogues (e.g., the dot-com crash, Global Financial Crisis, COVID-19, rate hike cycles, or AI boom).

Data Setup Tasks

- Get Bloomberg Access - **DONE**
- Get access to WRDS - **PENDING**
- Data Push: Pull at least 1 month of CRPS price data - **NOT DONE**
- API Key: Generate Claude & ChatGPT API Key - **DONE**
- Brainstorm Unstructured Data: List feasible unstructured sources (eg: news, earnings calls) - **DONE**

Brainstorming

Profile Agent Brainstorm

- **Goal:** Use Structured intake (career type, income stability, industry exposure, holdings, horizon, risk tolerance) to get total wealth assessment (human + financial capital)

a) Input & Data Retrieval Strategy

The system will process 3 explicitly provided core personas of biology Professor. tech executive. financial professional.

- **Structured data:** Financial holdings, age, investment horizon, target retirement age, and baseline risk tolerance.
- **Unstructured Inputs Human Capital Parameters:** Employment data, company name, job title, and geographic location.
- **Retrieval & Scalability Plan:**
 - **Baseline Personas:** Hardcode 3 provided personas into structured JSON profile files
 - **Synthetic Expansion:** Using Claude/ChatGPT API keys we shall **generate 30-50 randomized iterations of personas** (while varying factors like years to retirement, industry bonus structures & unvested stock options) to represent a complete client database.
 - **Open Labor Data APIs:** We can fetch aggregate income stability & wage data from U.S. Bureau of Labor Statistics (BLS) API or open-source data like Federal Reserve Survey of Consumer Finances (SCF). **Rationale:** This maps industry volatility metrics to personas without individual data scraping.

b) Output Schema (JSON)

Profile Agent shall output standardised schema to pass down the pipeline:

```
JSON
{
  "client_id": "synthetic_tech_exec_012",
  "financial_capital": 2000000,
  "human_capital_valuation": 4500000,
  "industry_exposure_sector": "Technology",
  "income_stability_score": "Low",
  "risk_tolerance_level": "Aggressive"
}
```

c) Failure Modes & Validation Plan

- **Failure Mode:** LLM hallucinates unrealistic human capital valuation
- **Validation Plan:** Implement deterministic sanity check layer in Python. If LLM's human capital valuation deviates significantly from actuarial formula based on work life expectancy, flag the profile for review.

Research Agent Brainstorm

- **Goal:** Use macro conditions from FRED and maps to historical market regime to get an idea of current market vulnerability.

a) *Input & Data Retrieval Strategy*

We need macroeconomic indicators to determine shifts in the economy.

- **Macro Variables:** Yield Curves (10Y minus 2Y Treasury), credit spreads, unemployment rates, CPI
- **Retrieval Plan:**
 - **FRED API:** Pull open-source macro series like **T10Y2Y** (Yield Curve), **UNRATE** (Unemployment), and **CPIAUCSL** (CPI).
 - **Unstructured Data:** Parse recent Federal Reserve meeting minutes or FOMC statements via RSS feeds to give the LLM qualitative sentiment parameters.

b) *Output Schema (JSON)*

JSON

```
{
  "current_macro_regime": "Rate_Hike_Cycle",
  "analogous_historical_regime": "2022_Inflation_Response",
  "regime_confidence_score": 0.85,
  "macro_indicators": {
    "yield_curve_status": "Inverted",
    "cpi_trend": "Increasing"
  }
}
```

c) *Failure Modes & Validation Plan*

- **Failure Mode:** LLM faces recency bias where it can label minor market corrections as systematic crises, or it hallucinates historical regimes that don't fit the data.
- **Validation Plan:** Hybrid approach. Perhaps run a clustering algorithm (eg: K-Nearest Neighbors) in Python directly on historic FRED macro data to identify the closest historical regime.
- LLM will be used to give an explanation with historical context.

[Data to download](#)

1. Wharton Research Data Services (WRDS)

A. CRSP (Center for Research in Security Prices)

- **Daily/Monthly Stock Returns:** Historical price, close price, shares outstanding, and total returns for individual equities spanning the specific regimes (Dot-com crash, GFC, COVID-19, 2022 rate hikes, and the AI boom).
- **Delisting Returns:** Critical to avoid survivorship bias in your backtesting models.
- **Market Indices:** S&P 500, NASDAQ, and Russell 3000 historical returns to benchmark your allocations against the standard \$80/20\$ benchmark.

B. Compustat (Fundamental Data)

- **Annual and Quarterly Financial Statements:** Balance sheets, income statements, and cash flow data (specifically Net Income, Total Debt, Cash Holdings, and Revenue growth).
- **Company Descriptors:** GICS Sector and Industry classification codes to map individual stocks directly to your client personas' industry exposures.

2. Bloomberg Terminal

A. Equity & Portfolio Risk Metrics

- **Beta & Equity Risk Factors:** Historical and predicted Betas (`BETA_RAW_OVERRIDE` / `BETA_ADJUSTED`) for individual stocks to feed into your Allocation and Risk agents.
- **Corporate Concentration Data:** Historical revenue segments by industry and geography for specific companies (crucial for evaluating a tech executive's employer-specific risk).

B. Fixed Income & Derivative Data

- **Yield & Bond Spreads:** Corporate bond yields across credit tiers (AAA down to High Yield) to construct fixed-income options for your portfolios.
- **Implied Volatility (IV):** Volatility smiles and index IV (VIX) history to assist your Research Agent with macro vulnerability scoring.

3. Federal Reserve Economic Data (FRED)

A. Yield Curve & Interest Rates

- **T10Y2Y:** 10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity (Yield Curve inversion tracker).
- **FEDFUNDS:** Effective Federal Funds Rate to identify rate-hike cycles.

B. Macroeconomic Health & Inflation

- **UNRATE:** Civilian Unemployment Rate to monitor labor market shocks.
- **CPIAUCSL:** Consumer Price Index for All Urban Consumers (Inflation tracking).
- **BAA10Y:** Moody's Seasoned Baa Corporate Bond Yield Relative to Yield on 10-Year Treasury Constant Maturity (Credit Spreads).

4. Labor & Unstructured Data Sources

A. Bureau of Labor Statistics (BLS)

- **Occupational Employment and Wage Statistics (OEWS):** Mean/median salaries and wage percentiles filtered by occupation codes (e.g., higher education professionals vs. technology managers).
- **Injuries, Illnesses, and Fatalities (IIF) / Unemployment by Industry:** To establish the empirical "income stability score" for each profession.

B. Qualitative Text Data (For LLM Ingestion)

- **Federal Reserve FOMC Minutes:** Plain text of recent Federal Reserve meeting statements to run sentiment parsing through your Research Agent.
- **Corporate Earnings Call Transcripts:** Transcripts of targeted companies (e.g., the tech executive's employer) to detect hidden concentration risks or forward-looking industry guidance.

Workflow

Step 1: Gather Foundation Data (WRDS & FRED)

We need to get an idea of what we can use

- **Why?** The baseline macro indicators (from FRED) and historical stock lists/returns (from WRDS) to define your 4–5 specific market regimes (like the GFC or the AI boom)
- **Expected Result:** Clear Mapping of which years shows which regimes and universe actual tickers/returns to allocate assets into.

Step 2: Establish the 3 Core Baseline Personas

We need to hardcode exact profiles given the 3 provided personas (biology professor, tech executive & finance professional) using realistic parameters based on labour data (BLS).

Step 3: Programmatically Generate 30-50 iterations

- Use Claude/ChatGPT to generate synthetic profiles based on real-world data. Eg: When creating a variant of Tech Executive, we can programmatically restrict holdings to actual tech stocks pulled from WRDS in step 1.